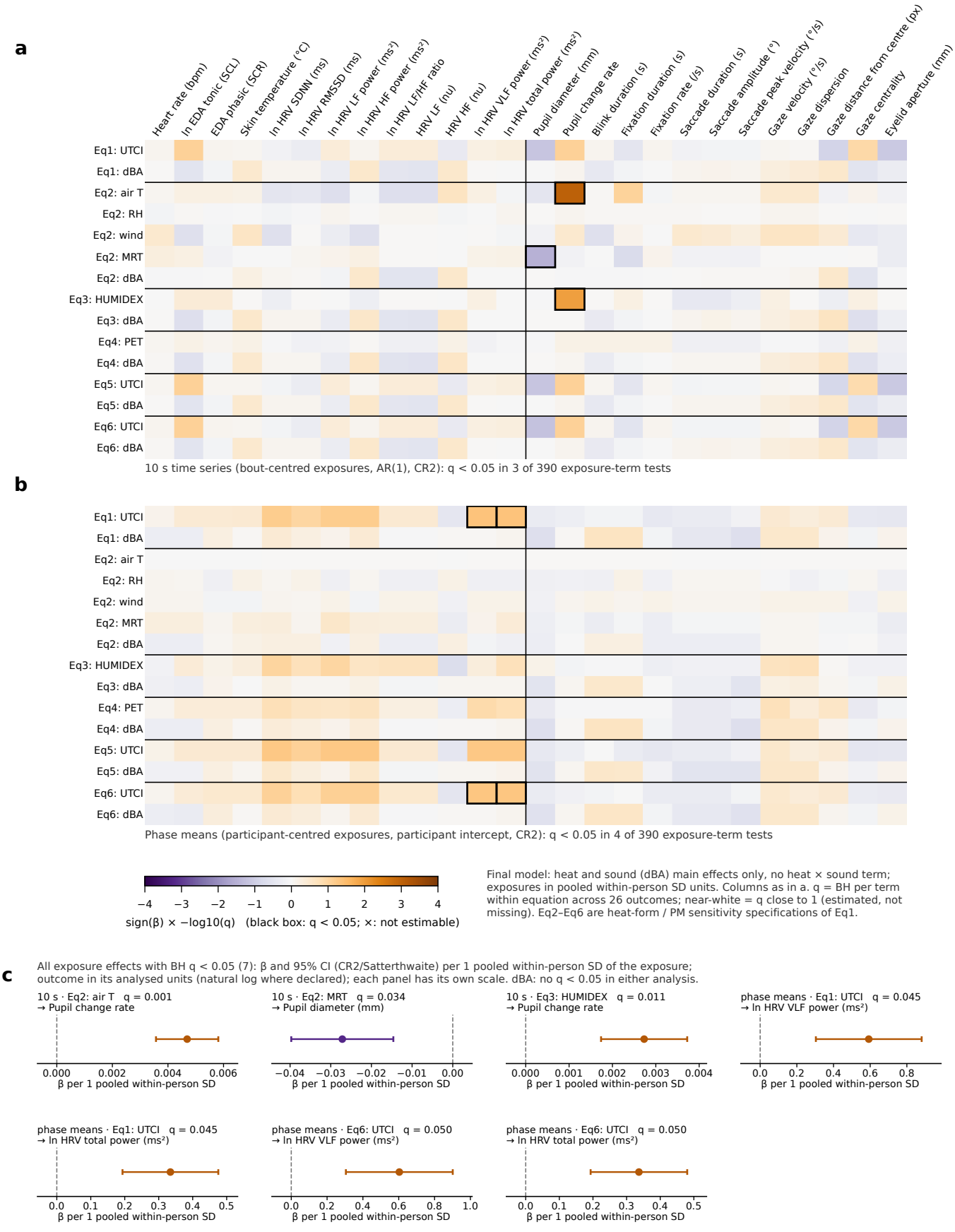
a
Heart rate (bpm)
ln EDA tonic (SCL)
EDA phasic (SCR)
Skin temperature (°C)
ln HRV SDNN (ms)
ln HRV RMSSD (ms)
ln HRV LF power (ms²)
ln HRV HF power (ms²)
ln HRV LF/HF ratio
HRV LF (nu)
HRV HF (nu)
ln HRV VLF power (ms²)
ln HRV total power (ms²)
Pupil diameter (mm)
Pupil change rate
Blink duration (s)
Fixation duration (s)
Fixation rate (/s)
Saccade duration (s)
Saccade amplitude (°)
Saccade peak velocity (°/s)
Gaze velocity (°/s)
Gaze dispersion
Gaze distance from centre (px)
Gaze centrality
Eyelid aperture (mm)
Eq1: UTCI
Eq1: dBA
Eq2: air T
Eq2: RH
Eq2: wind
Eq2: MRT
Eq2: dBA
Eq3: HUMIDEX
Eq3: dBA
Eq4: PET
Eq4: dBA
Eq5: UTCI
Eq5: dBA
Eq6: UTCI
Eq6: dBA
10 s time series (bout-centred exposures, AR(1), CR2): q < 0.05 in 3 of 390 exposure-term tests
b
Phase means (participant-centred exposures, participant intercept, CR2): q < 0.05 in 4 of 390 exposure-term tests
−4 −3 −2 −1 0 1 2 3 4
sign(β) × −log10(q) (black box: q < 0.05; ×: not estimable)
Final model: heat and sound (dBA) main effects only, no heat × sound term; exposures in pooled within-person SD units. Columns as in a. q = BH per term within equation across 26 outcomes; near-white = q close to 1 (estimated, not missing). Eq2–Eq6 are heat-form / PM sensitivity specifications of Eq1.
c
All exposure effects with BH q < 0.05 (7): β and 95% CI (CR2/Satterthwaite) per 1 pooled within-person SD of the exposure; outcome in its analysed units (natural log where declared); each panel has its own scale. dBA: no q < 0.05 in either analysis.
10 s · Eq2: air T q = 0.001
→ Pupil change rate
0.000 0.002 0.004 0.006
β per 1 pooled within-person SD
10 s · Eq2: MRT q = 0.034
→ Pupil diameter (mm)
−0.04 −0.03 −0.02 −0.01 0.00
β per 1 pooled within-person SD
10 s · Eq3: HUMIDEX q = 0.011
→ Pupil change rate
0.000 0.001 0.002 0.003 0.004
β per 1 pooled within-person SD
phase means · Eq1: UTCI q = 0.045
→ ln HRV VLF power (ms²)
0.0 0.2 0.4 0.6 0.8
β per 1 pooled within-person SD
phase means · Eq1: UTCI q = 0.045
→ ln HRV total power (ms²)
0.0 0.1 0.2 0.3 0.4 0.5
β per 1 pooled within-person SD
phase means · Eq6: UTCI q = 0.050
→ ln HRV VLF power (ms²)
0.0 0.2 0.4 0.6 0.8 1.0
β per 1 pooled within-person SD
phase means · Eq6: UTCI q = 0.050
→ ln HRV total power (ms²)
0.0 0.1 0.2 0.3 0.4 0.5
β per 1 pooled within-person SD